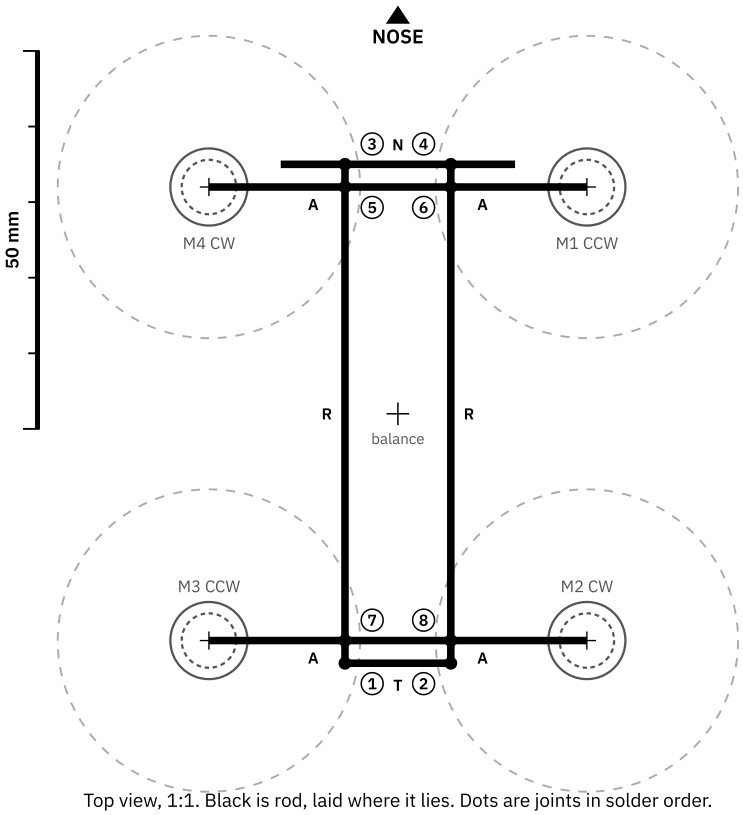
XS drone frame jig

Print on US Letter at Actual size (100 %), never Fit to page.
Both bars must read 50 mm on a caliper before you cut or lay anything.

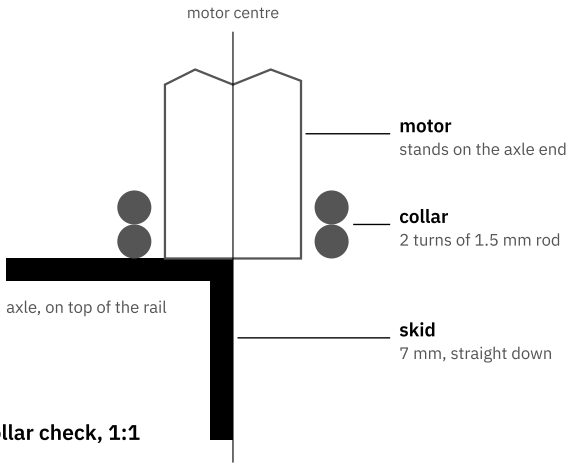
50 mm



Top view, 1:1. Black is rod, laid where it lies. Dots are joints in solder order.
Grey rings are the collars, dashed circles the 40 mm propellers.

Axle end, side view, 3 × size

Not for laying rod: the 1:1 view is on the left.



Collar check, 1:1



inside 7.2 mm, outside 10.2 mm
A wound collar should cover the grey ring.

Joints

Laps: each cross piece lies on top of the rails.
Collars sit on the axle ends; skids hang under them.
The battery straps between the rails, nose to tail;
the frame stays electrically floating.

Cut list, 1:1. Lay the rod on its bar and mark it at the end tick.

R rails · 2 × 66 mm · 1 mm rod

A axles · 2 × 50 mm · 1 mm rod

N nose bar · 1 × 31 mm · 1 mm rod

T tail bar · 1 × 14 mm · 1 mm rod

S skids · 4 × 7 mm · 1 mm rod

C collar blanks · 4 × 66 mm · 1.5 mm rod

The ten 1 mm pieces take 305 mm plus saw cuts: two 300 mm brass rods (the axles from the second). The four 66 mm collar blanks come from one 300 mm rod of 1.5 mm brass.

Build order

1. Test first. On two offcuts of 1.5 mm brass, wind one collar and solder one lap and one T with rosin flux and 60/40 tin. Bend them hard. A joint that pops off clean means more heat before the frame.
2. Cut. Lay each rod on its bar below, mark the end tick, saw 0.5 mm long and sand square to the mark.
3. Set up. Tape this sheet to the silicone mat (never the cutting mat). Put a Kapton square over each numbered dot, then hold the rails on their lines with Kapton strips.
4. Solder 1 to 4: tail bar, then nose bar, both laid across the tops of the rails. Check the rail diagonals: 67.5 mm both ways.
5. Solder 5 to 8: both axles across the tops of the rails, each end on its motor centre.
6. Collars. Set one on each axle end, lower turn on the axle, with the 7 mm former (a drill shank) left inside so it stays round. Solder, then pull the former out. A 7 × 20 motor should push in by hand.
7. Flip the frame and solder a 7 mm skid under each axle end, straight down.
8. Brass needs no washing: wipe the rosin off with alcohol, then weigh: about 5.3 g.